Deep or Shallow Facial Descriptors? A Case for Facial Attribute Classification and Face Retrieval

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Abstract. With the largely growing quantity of face images in the social networks and media, different face analyzing systems are developed to be employed in real-world situations such as face recognition, facial expression detection, or automated face tagging. Two demanding face-related applications are studied in this paper: facial attribute classification and face image retrieval. The main common issue with most of the attribute classifiers and face retrieval systems is that they fail to perform well under various facial expressions, pose variations, geometrical deformation, and photometric alterations. On one hand, the emerging role of deep CNNs (convolutional neural networks) has shown superior results in tasks like object recognition, face recognition, etc. On the other hand, their applications are yet to be more investigated in facial attribute classification and face retrieval. In this study, we compare the performance of shallow and deep facial descriptors in the two mentioned applications by proposing to exploit distinctive facial features from a very deep pre-trained CNN for attribute classification as well as constructing deep attribute-driven feature vectors for face retrieval. According to the results, the higher accuracy of the attribute classifiers and superior performance of the face retrieval system is demonstrated.

1 Introduction

Recent developments in social networks and medias have heightened the need for analyzing images because of the important applications they have in real-world contexts such as large-scale image retrieval [1, 2], automated image annotation [3, 4], and scene understanding [5]. Millions of images are uploaded to social networks every day, Facebook, for instance, process approximately 350 million faces daily [6]. Of all these images, 45% are estimated to encompasses faces [7] highlighting the importance of processing faces for purposes such as face recognition [8], facial expression detection [9], or face annotation [10]. However, retrieving similar face images using a single query face image remains a challenging problem in many real-world applications, particularly in law enforcement investigations

Fig. 1. Sample faces of one person under different expressions, head poses, facial occlusions, geometrical deformations, and photometric alterations.

where it assists authorities with narrowing leads and, subsequently, confirming a suspects identity. The goal of such content-based Face Image Retrieval (FIR) systems is to retrieve top most similar faces from a large database faces (distracting or gallery). Such databases are challenging due to the presence of diverse face images from multiple persons. There are also large intra-class variations due to facial expressions, pose variations, geometrical deformation, and photometric alterations (Fig. 1). Hence, FIR has been an active field of re-search among the scholars over the recent years [11–18].

In order to foster a new attribute-driven FIR system to fill in the gap of the existing research, this study will examine the emerging role of very deep pre-trained convolutional neural net-work (CNN) [8] in the context of supervised semantic facial attribute classification and, more importantly, its impact on performance, mean average precision (mAP), of FIR systems whilst using attribute-driven feature vectors. Our research contributions are listed as:

- Semantic facial attribute classifiers are trained by exploiting salient facial features from the last fully connected layer of a very deep pre-defined CNN (VGG-Face) [8] which is designed and trained for the task of face recognition. Although the quantity of training faces is not large, the accuracy is higher than shallow-based attribute classifiers.
- We compared the accuracy of the proposed deep descriptor classifiers with those of the shallow descriptors (D-SIFT, G-LBP, and G-HOG) to investigate their effectiveness regarding prediction of each facial attributes (eight attributes). This is the only comparative study, to the best of our knowledge, com-paring deep and shallow features in the context of face attribute classification.
- The mean average precision of (mAP) shallow attribute-driven face image retrieval is increased by more than double by exploiting deep attribute-driven

facial features resulted from the trained binary classifiers. We compared the performance (mAP) of proposed deep attribute-driven FIR with the performance of shallow attribute-driven FIRs with respect to different gallery (distracting) face database sizes (2k, 4k, 6k, 8k, and 10k) constructed using faces from CelebA database [19].

This paper is organized as follows. The next section briefly discusses recent related works in face attribute classification and face image retrieval. Section 3 presents the methodology of our proposed system while section 4 and 5 discusses the experimental results and discussion. It is followed by section 6 which is conclusion of the paper.

2 Related Work

This research is tightly related to two research areas, comprising facial attribute classification, and face image retrieval (FIR). The conventional facial attribute classifiers extract low-level features (SIFT, LBP, HOG, etc.) within different localized face components (two eyes, nose, and two mouth corners) from a given face and feed them to a supervised classifier such as linear SVM [20]. Following this approach, along with extracting features from overlapped components, they tolerate some degrees of error in delocalized face components [15]. Recently, the studies in [21] and [19] extracted features from deep CNNs aiming for training more accurate face attribute classifiers. In [21] the CNN architecture in [8] was adopted and the cost function was replaced with a new proposed one. The new architecture was trained using 160k face images from CelebA [19] database and it took 2,500 iterations to a complete one epoch on the collection of training faces. The second work presented in [19] used two pre-trained CNNs, known as LNet, and ANet, to fine-tune them for tasks of face localization and attribute classification, respectively. To train LNet and ANet, 1.2 million images of 1k distinct objects, and 160k face images of 8k different identities were employed, respectively. However, fine-tuning a pre-trained deep CNN or training a newly designed one is very resource-demanding (a large corpus of training images is required) and computationally heavy, as with the works in [19, 21]. These two studies, to date, are the only ones exploiting deep CNNs for the task of supervised facial attribute classification.

To increase the performance of FIR systems several methods have been proposed recently. In [16], to give distinctiveness to the facial features, soft biometrics along with facial marks (moles, scars, etc.) were taken into account to increase the FIR precision. An automated facial mark detection was proposed for this purpose too. The semi-supervised FIR method in [22] was the first to introduce the application of sparse-coded features constructed from locally extracted features. This work achieved scalability by proposing a novel coding scheme and combining inverted indexing and sparse features. The work in [15] was the first introducing a multi-reference re-ranking and identity quantization model to achieve scalability and precision while retrieving face from a database

of 1 million images. An unsupervised FIR method based on voting scheme and classification was introduced in [11]. The faces are represented using local features and then, to speed computations, an approximate nearest neighbor (ANN) search is used to find matching features. Another work in [12] attempted to take advantage of a deep neural net (NN) to pre-filter the database faces using extracted features from the learned NN. A k-NN search was employed to find the top-k most similar faces while retrieving.

The state-of-the-art FIR systems achieve scalability and precision by exploiting both high-level (semantic) and low-level facial attributes [17, 18]. They first retrieve the N top similar faces using semantic attribute-driven features, then the re-rank them, to get the top M (M¡N) relevant faces (e.g. top 10), by their low-level facial descriptors extracted from sub-regions (patches) within different localized face regions (components) [15]. To date, the only attribute-driven FIR techniques are those proposed in [17, 18, 23].

3 Method

An overall pipeline of our FIR approach is presented in Fig. 2. Given a face image as a query, a database of gallery faces (distracting), and a database of training faces, the system begins by extracting the deep facial features using a very deep pre-trained CNN, followed by training eight supervised binary classifiers for eight facial attributes (Fig. 3). Then, extracted deep facial features from the query face and gallery sets are fed to the eight trained classifiers in the previous stage in order to compute the Fuzzy attribute scores of each unique attribute. Finally, the attribute scores calculated for each face image are concatenated to construct a deep attribute-driven facial feature vector for that face. The procedure for the last step is illustrated in Fig. 4.

3.1 VGG-Face – A Very Deep Pre-trained CNN

In our proposed method, we extract facial features from the very deep pre-trained CNN (VGG-Face) [8] which is designed and trained for the task of face recognition. This architecture is referred to as very deep as it comprises a long sequence of convolutional layers. The VGG-Face has 37 layers and is was trained by a triplet-loss method using 2.6 million faces (2,633 identities) collected from the wild. To solve the problem of recognizing N=2,633 distinctive identities, the deep CNN architectures were bootstrapped (N-ways classification). Then, each training image was associated with a training score vector considering the last fully-connected layer including N linear predictors. Finally, all scores were compared to the ground-truth by calculating the statistical soft-max log-loss and their scores were further improved by triplet-loss tuning.

3.2 Facial Feature Extraction

For each face image, four 224×224 patches are cropped (from the center with horizontal flip and four corners) and an averaged feature vector is constructed

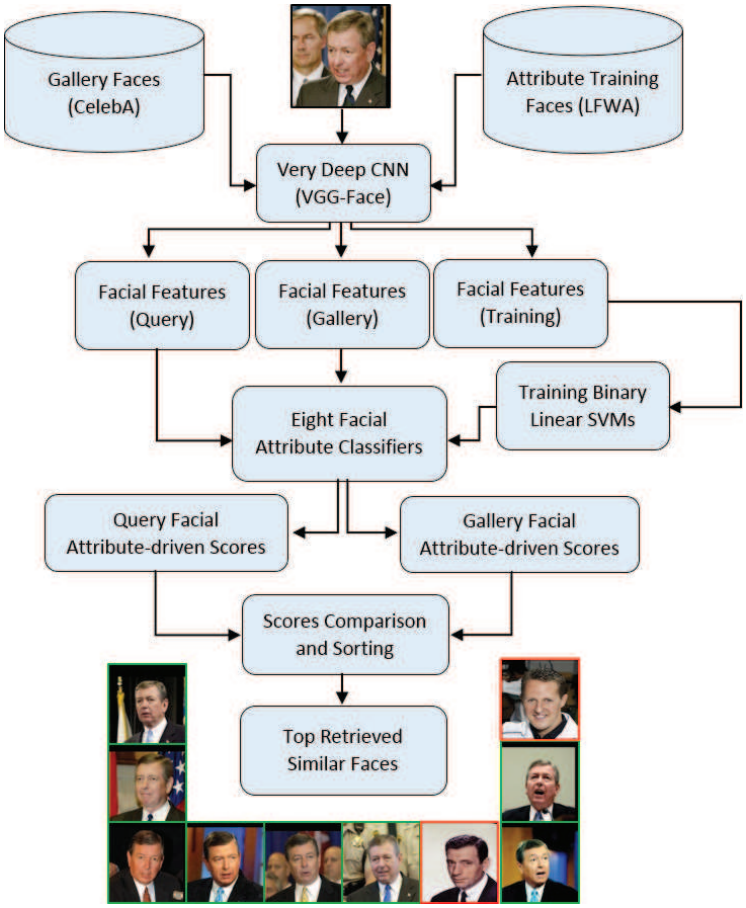

Fig. 2. The overall system level diagram of the proposed FIR system: faces in green boxes are true positives, and those in red boxes indicate false positives.

based on them. In order to take advantage of multi-scale testing, the given face is scaled to different sizes of 512, 384, and 256 pixels followed by replicating the cropping procedure for all of them. To extract the final facial features, the 4,096-dimensional vector is extracted from the last fully-connected layer in the VGG-Face which is the results of averaging all feature vectors.

3.3 Binary Linear SVMs

For the attribute classification task, features are extracted from both training and test images. Both set of images contain positive and negative samples from the corresponding attribute. Eight linear binary SVM classifiers are trained and tested on the eight facial attributes classification tasks.

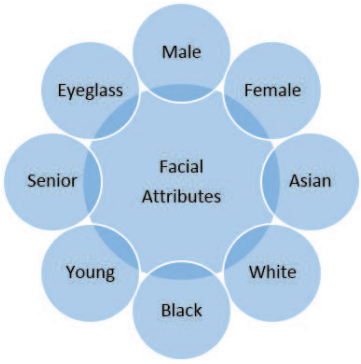

Fig. 3. List of eight facial attributes employed in this research.

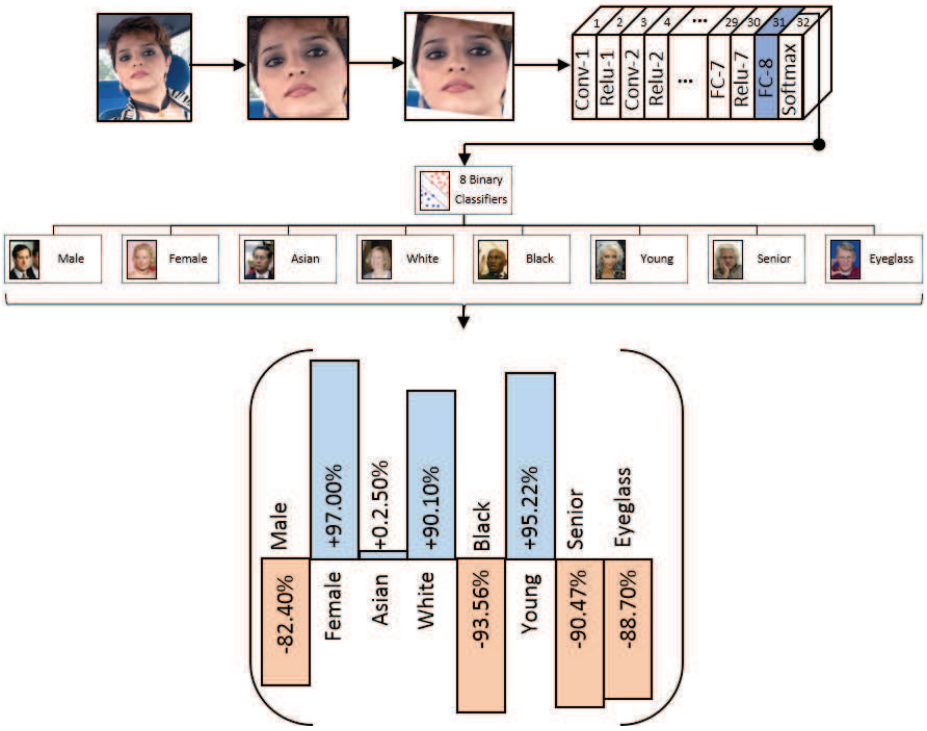

Fig. 4. Construction of deep attribute-driven facial feature vectors

The choice and the number of positive and negative samples from LFW [24] dataset are listed in Table 1.

Table 1. Types of facial attribute classifiers and their choice and number of training face images selected from LFW face dataset.

Type of Classifiers	Selection of Positive and Negative Face Images	# Training and Testing Faces	Sample Faces used for Training
Male Classifier	Male vs. Female	2,000 – 2,000	
Female Classifier	Female vs. Male	2,000 – 2,000	
Asian Classifier	Asian vs. (Black + White)	1,000 – 1,000	
White Classifier	White vs. (Asian + Black)	1,000 – 1,000	
Black Classifier	Black vs.(Asian + White)	500 - 500	
Young Classifier	Young vs. Senior	2,000 – 2,000	
Senior Classifier	Senior vs. Young	2,000 – 2,000	
Eyeglass Classifier	Eyeglass vs. No Eyeglass	2,000 – 2,000	

3.4 Attribute-driven Fuzzy Scores

In this part, the 4,096 dimensional facial feature vectors (from Section 3.1) extracted from the query face and all the faces in the gallery database are used as inputs to the eight attribute classifiers constructed in the previous section. Then, all the resulted attribute scores for each face image are L2 normalized and concatenated to build the attribute-driven Fuzzy scores for that face (Fig. 4).

3.5 Scores Comparison and Sorting

Finally, by comparing the Euclidean distance between the attribute-driven Fuzzy scores of the query face and all the faces in the gallery database, we sort the faces in the gallery from the minimum to maximum. The first top N similar faces are the first N in the final sorted list.

4 Experiments

In this section, two sets of experiments are conducted to evaluate the performance of the attribute classifiers using deep descriptor and deep attribute-driven FIR system with respect to different database sizes ranging from 2k to 10k face images.

Table 2. Names of the individuals in the LFW used as query set and their associated numbers of faces used in query and gallery sets.

No	Person Name	Sample Face	# of Query Faces	# of Faces in the gallery	Total number of faces
1	Ariel Sharon		10	67	77
2	Colin Powell		10	226	236
3	Donald Rumsfeld		10	111	121
4	George W Bush		10	520	530
5	Gerhard Schroeder		10	99	109
6	Hugo Chavez		10	61	71
7	Jacques Chirac		10	42	52
8	Jean Chretien		10	45	55
9	John Ashcroft		10	43	53
10	Junichiro Koizumi		10	50	60
11	Serena Williams		10	42	52
12	Tony Blair		10	134	144

4.1 Databases

The LFW (labeled faces in the wild) [24] is a face image dataset developed for studying the problem of unconstrained face recognition. The database includes over 13,000 images of faces collected from the WWW. From LFW database, the training faces are selected as detailed in Table 1. Following the experiments performed in [17, 18], we select 12 unique individuals from LFW database with more than 50 face images per person (totally 1,560 faces), from these collection, we selected 10 face images for each individual (12 individuals) to be served as our query faces (totally 120 faces). Samples and numbers of selected faces for both quay and gallery sets are shown in Table 2. Using Viola-Jones face detector [25], all the faces are detected from the entire collection of face images and then they are resized to 224×224 using bilinear interpolation method.

CelebA [19] dataset (Celebrity Faces Attributes) is a scalable face image dataset including over 200k faces of celebrities (over 10k distinct people). From this dataset, we select five subsets of 2k, 4k, 6k, 8k and 10k faces to be served as our gallery databases while performing face retrieval. There is no overlap between the faces in the training set and those relevant faces in the gallery databases.

4.2 Performance Evaluation and Comparison

For face retrieval, mAP at 100% recall is used as performance measurement. The mAP is defined as the mean of all average precisions resulted from inquiring each unique face query image.

In order to compare the performance of deep facial descriptors with the shallow descriptors in the contexts of attribute classification and face retrievals, three feature descriptors are selected as Histogram of Oriented Gradients (HOG) [26] which its successfully used in face recognition high accuracy [27], Local Binary Pattern (LBP) [28] which has proved to work well in face recognition tasks as reported in [29], and Scale Invariant Feature Transform (SIFT) [30].

4.3 Parameter Settings

To facilitate the procedure of re-implementing the proposed methodology, here we report the parameter settings of the binary linear SVM classifiers and shallow feature extractors. For the SVMs a linear kernel function was used and SMO (Sequential Minimal Optimization) method was selected to find the separating hyperplanes. Two input parameters for the kernel MLP (Multilayer Perceptron) are set as 1 and -1. To extract the HOG, LBP, and SIFT features we used the grid version (G-HOG, G-LBP, and G-SIFT) with the grid step of 8×8 and block width of 32×32.

4.4 Software and Hardware

For G-SIFT extraction, VLFeat Toolbox (0.9.20) [31] was used, and other programming parts are done using Matlab R2016a. Our experiments are performed on a desktop computer with Windows 7 (64 bit), 16 GB RAM, Intel Core i7-2600 CPU @ 3.40GHz.

5 Results and Discussion

In this section, the accuracy of eight trained binary attribute classifiers, as well as mAP of the proposed FIR system, are presented. Furthermore, recommendations on future works are founded on analysis of the results.

5.1 Facial Attribute Classification Performance

Based on the accuracies reported for each attribute in the Fig. 5, performance of deep facial descriptors is always better than the shallow descriptors, but of the shallow ones, G-SIFT is comparable with the highest (achieved by deep descriptors). Regarding the young and senior classification accuracies, their difference between G-SIFT and Deep CNN is inconsiderable, it is 0.45% and 1.05% for the young and senior classification (age category), respectively. On the other hand, the biggest difference can be seen in the Asian, white, and black classifications

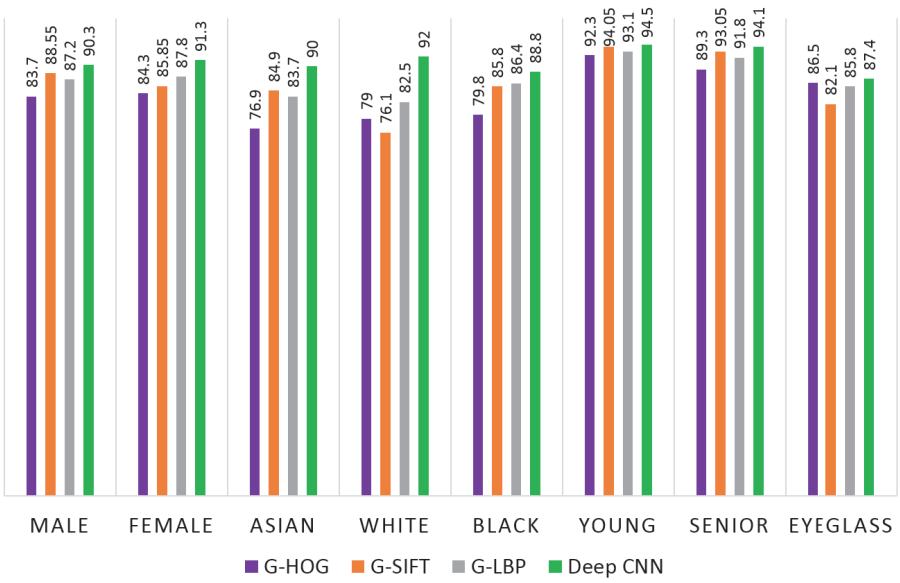

Fig. 5. Accuracy of various facial attribute classifiers. Accuracies using three types of shallow descriptors (G-HOG, G-LBP, and D-SIFT) and one deep descriptor.

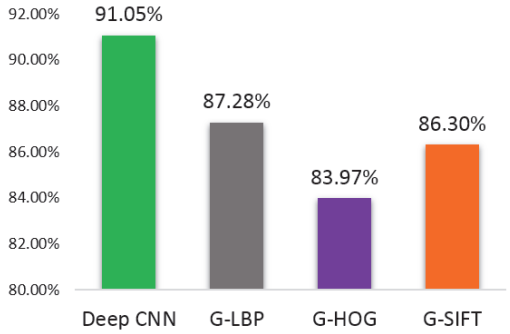

Fig. 6. Overall performance of deep and shallow facial descriptors in the task of attribute classification..

(race category). The performance of G-LBP is comparable with deep CNN in the eyeglass classification while the difference is 1.6%. Finally, the accuracy of deep descriptors is higher than the shallow descriptors in the gender classification task (male and female). Overall, of the shallow facial descriptors, G-LBP has the highest classification accuracy and the lowest is given by G-HOG (Fig. 6). The total performance difference between deep descriptors and the best shallow descriptor (G-LBP) accounts for less than 4%.

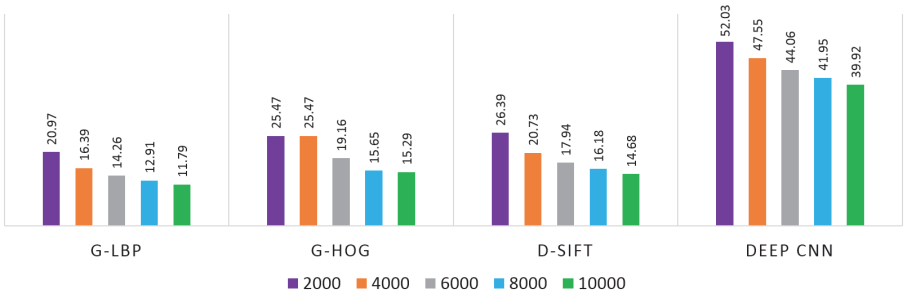

Fig. 7. Mean average precision (mAP) of different FIR systems using shallow and deep attribute-driven feature vectors. Results are based on different database sizes.

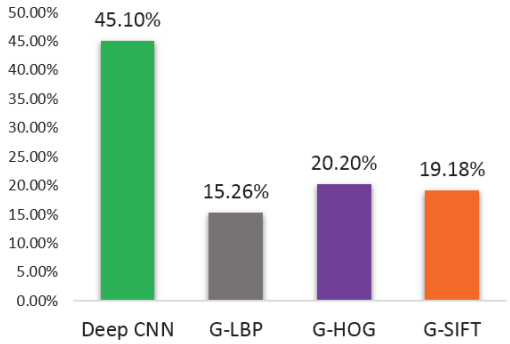

Fig. 8. Overall performance of deep and shallow facial descriptors for face image retrieval task.

5.2 Face Image Retrieval Performance

According to the mAPs reported in the Fig. 7, the performance of deep attribute-driven FIR is always superior to the shallow attribute-driven FIR systems by 24.63% increase (more than double) in the context of diverse face datasets. Regardless of the database size, the mAP of the deep attribute-driven FIR is always remarkably higher than the shallow-based FIR systems making them incomparable. Overall, among all of the retrieval performances attained by shallow descriptors, G-HOG has the best at 20.20% which is only 1.02% higher than G-SIFT (Fig. 6). Sample retrieved faces and their associated query faces are illustrated in Fig.9 .

5.3 Discussion and Recommendation

Overall, it can be inferred that shallow facial descriptors used in this study (G-HOG, G-LBP, and G-SIFT) are still capable of classifying eight facial attributes (Fig. 3) with a slight difference compared to the deep facial descriptors, but their

Fig. 9. Examples of the top-10 retrieved face images in the proposed FIR system. Query images are given 3in blue boxes. The faces are ranked from left to right, and top to bottom. True positives are indicated by green boxes while false positives by red boxes.

performances largely degraded in the task of attribute-driven FIR as opposed to the deep facial descriptors. Regarding shallow descriptors, while the G-LBP achieved the best performance for attribute classification, the worst retrieval performance was attained by the same descriptor. Conversely, G-HOG has the best performance for the face retrieval task but it also performed the worst in attribute classification. This highlights the fact that handcrafted shallow facial descriptors are often tailored to perform well in certain applications only, as it does not learn the best possible set of features in a natural way.

Despite the superior performance of deep facial descriptors in both face retrieval and facial attribute classification, they are computationally intensive and time-consuming. In the case of G-SIFT (as stated in Fig. 10), the speed of both feature extraction and classification are more than double in contrast with the deep CNN descriptors. The speed is remarkably higher for G-LBP and G-HOG with more than two orders of magnitude (10^2) faster than generating the deep descriptors.

Therefore, we recommended that future research in the field of facial attribute classification should attempt to focus more on designing CNNs that are able to achieve higher accuracy while decreasing the execution time for both feature extraction and classification. As most state-of-the-art research in this field exploit a large number of training (positive and negative) faces to achieve good results, CNNs are most suited to provide the necessary generalization.

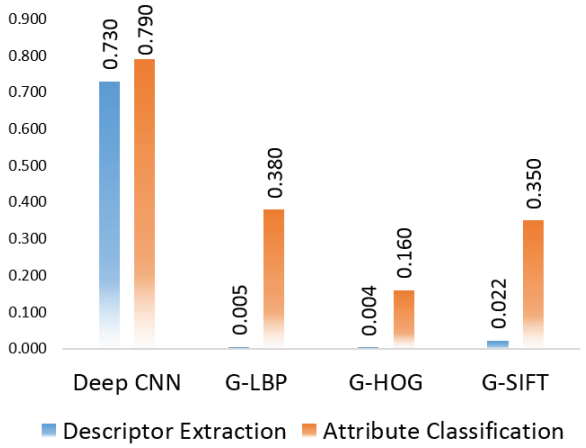

Fig. 10. Average execution time for the both descriptor extraction and attribute classification according to deep and shallow descriptors.

With respect to an attribute-driven face retrieval system, as the retrieval performance is correlated with the performance of facial attribute classification. Thus, it is essential to increase the number of facial attributes used to describe faces while utilizing more accurate attribute classifiers. In future, we intend to design an attribute-specific CNN architecture for face retrieval that is both efficient (small number of network parameters) and effective (maximizing the accuracy).

6 Conclusion

In this paper, we exploited deeply learned facial descriptors to increase the accuracy of face attribute classifiers compared to conventional shallow descriptors (HOG, LBP, and SIFT). We also utilized deep attribute-driven facial feature vectors to enhance the performance of an FIR system. Although the accuracy of the deep attribute classifier is comparable to that of shallow attribute classifiers, its performance in the face retrieval task increased by more than double. This demonstrates the capability of deeply learned facial descriptors in improving the performance of an FIR system.

In this preliminary research, we have used a reasonable amount of face images (10k maximum) to train a reduced set of attribute classifiers, owing to computational limitations. Hence in future, we intend to further validate this work on larger number of faces from the selected datasets (particularly the large-scale CelebA dataset). Besides, a larger number of attributes will also be used to provide more discrimination power to the deep attribute-driven feature vectors while scoring and retrieving similar-appearing faces. Also, we intend to use fea-

tures from localized face components and to further investigate their effect on the overall performance of both tasks.

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